Yuchen Shen

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Education

Carnegie Mellon University 2025/08-present
Ph.D. in Machine Learning, Machine Learning Department Pittsburgh, PA, USA
Carnegie Mellon University 2023/08-2024/12
Master of Science in Intelligent Information Systems, Language Technologies Institute Pittsburgh, PA, USA
University of Electronic Science and Technology of China 2019/08-2023/06
Bachelor of Engineering in Software Engineering, School of Information and Software Engineering Chengdu, China

Publications

* Equal contribution.

[1] Jeremy Tien, Abishek Anand, Yu-Rou Tuan, **Yuchen Shen**, J. Zico Kolter, Aran Nayebi. ROGUE: Evaluating Corrigibility Failures in Frontier Computer-Use Agents. **arXiv preprint, 2026**.
[2] Trinity Chung*, **Yuchen Shen***, Nathan C. L. Kong, Aran Nayebi. Task-Optimized Convolutional Recurrent Networks Align with Tactile Processing in the Rodent Brain. **NeurIPS 2025 (Oral)**.
[3] **Yuchen Shen**, Haomin Wen, Leman Akoglu. FoMo-0D: A Foundation Model for Zero-shot Tabular Outlier Detection. **TMLR 2025**.
[4] **Yuchen Shen***, Chenhao Zhang*, Sijie Fu*, Chenghui Zhou, Newell Washburn, Barnabás Póczos. Chemistry-Inspired Diffusion with Non-Differentiable Guidance. **ICLR 2025**.
[5] Zhichao Hou, Weizhi Gao, **Yuchen Shen**, Feiyi Wang, Xiaorui Liu. ProTransformer: Robustify Transformers via Plug-and-Play Paradigm. **NeurIPS 2024**.
[6] **Yuchen Shen**, Barnabás Póczos. GraphBPE: Molecular Graphs Meet Byte-Pair Encoding. **ICML 2024 AI for Science Workshop**.
[7] **Yuchen Shen**, Xiaojun Wan. OpinSummEval: Revisiting Automated Evaluation for Opinion Summarization. **arXiv preprint, 2023**.
[8] Fei Zhao*, **Yuchen Shen***, Zhen Wu, Xinyu Dai. Label-Driven Denoising Framework for Multi-Label Few-Shot Aspect Category Detection. **Findings of EMNLP 2022**.
[9] Mengjuan Liu, Xiaoming Bao, Jiang Liu, Pei Zhao, **Yuchen Shen**. Generating emotional response by conditional variational auto-encoder in open-domain dialogue system. **Neurocomputing**, 460:106-116, 2021.

Selected Research

ROGUE: Agent Safety and Corrigibility, Carnegie Mellon University 2026 preprint
Advisor: Aran Nayebi.

- Coauthored ROGUE, an 82-task benchmark evaluating whether computer-use agents respect human intervention, shutdown, and access restrictions during ordinary tasks.
- The study identifies gaps between text-only and agentic safety evaluations and examines failures to preserve safety constraints when agents delegate to subagents.
- Received a **$6,000 Corrigibility Research Fund team prize**, September 2026.

FoMo-0D: Zero-Shot Tabular Outlier Detection, Carnegie Mellon University 2024/05 – 2024/09
Advisor: Leman Akoglu.

- Developed a prior-data fitted network pretrained on 1.6 million synthetic datasets for zero-shot tabular outlier detection, requiring no task-specific fine-tuning or labeled data.
- Outperformed most of 26 baselines across 57 real-world datasets; averaged 7.7 ms inference per sample in the reported experiments. Published in **TMLR 2025**.

Research Experience

Task-Optimized Tactile Models with Feedback, Carnegie Mellon University 2024/09 – 2025/05
Advisor: Aran Nayebi.

- Developed **PyTorchTNN**, a Python package that unrolls neural networks over time with feedback for biologically inspired modeling.
- Co-first-authored a study of recurrent tactile models and self-supervised learning objectives that align model representations with rodent neural responses. **NeurIPS 2025 Oral**.

ChemGuide: Diffusion with Non-Differentiable Guidance, Carnegie Mellon University 2024/01 – 2024/05
Advisor: Barnabás Póczos.

- Developed zeroth-order gradient estimation from non-differentiable quantum-chemistry oracles to guide pretrained diffusion models for 3D molecule generation.
- Combined molecular stability and property objectives in a bilevel guidance framework; evaluated improvements on QM9 and GEOM. Co-first-authored **ICLR 2025** paper.

Graph Tokenization for Molecules, Carnegie Mellon University 2024/01 – 2024/05
Advisor: Barnabás Póczos.

- Generalized byte-pair encoding to molecular graphs, developing a non-parametric subgraph tokenization method and evaluating its effects on graph and hypergraph neural networks.

Convergence of Decentralized Optimization Algorithms, North Carolina State University 2023/02 – 2023/12
Advisor: Xiaorui Liu.

- Analyzed convergence rates and lower bounds for decentralized algorithms under convex and strongly convex objectives.

Automated Metric Evaluation for Opinion Summarization, Peking University 2022/12 – 2023/05
Advisor: Xiaojun Wan.

- Constructed a human-judgment dataset covering outputs from 14 opinion-summarization models; compared 24 automated metrics across aspect relevance, coherence, sentiment consistency, and readability.
- Analyzed where automated metrics fail to track human judgments, including limitations of metrics built on strong language-model backbones.

Label-Driven Denoising for Few-Shot Aspect Detection, Nanjing University 2021/08 – 2022/01
Advisor: Xinyu Dai.

- Designed label-guided attention and a label-weighted contrastive loss to reduce noisy features and distinguish semantically close aspect categories.
- Improved Proto-AWATT F1 from 75.37 to 78.27 on FewAsp in the 5-way, 5-shot setting. Co-first-authored **Findings of EMNLP 2022** paper.

Skills

Language: Chinese (native), English (fluent) **Programming**: PyTorch/Python (proficient), C, Java
Research methods: Agent evaluation, synthetic-data pretraining, diffusion models, recurrent networks, optimization